CSE 321– Operating Systems

QUIZ 3

Summer– 2022

Grade

Full Name (in Block Letter): ____

ID: ______________________ Section:________ Signature: __________________

Marks: 15 points **Time: 30 minutes**

<u>Instructions:</u> Answer all questions on the space provided below for each.

Question 1 CO5 (2+3+3points):
Suppose, we have the following snapshot in a system. There are <u>four</u> processes and <u>four</u> resource types. Answer the following questions using Banker's Algorithm.

Processes	Max				Allocation				Available			
	A	B	C	D	A	B	C	D	A	B	C	D
Pa	7	0	1	3	7	0	0	2	1	4	4	4
Pb	2	7	5	0	2	1	0	0				
Pc	2	12	5	6	0	6	3	3				
Pd	1	6	5	6	0	2	1	2				

a. Calculate Need matrix.
b. Is this system in safe state? If yes, then find the safe sequence or if not, then provide necessary explainations?
c. What happens if process Pb request at this moment for (0, 4, 2, 0)? Whether Banker's algorithm grants the request or not? If grants the request find the safe sequence.

Question 2 CO5 (4+3points):

a. **Show an example and explain the necessary conditions for a deadlock to occur?**
b. **How can you prevent Deadlock? How can you eliminate circular wait?**